

Day 10—LINUX-FILES

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THEORY

What is a Linux directory?

It is a folder used to organize and store files and other directories.

What does mkdir do?

It creates a new directory (folder).

What does touch do?

Touch creates a file and it can edit a file timestamps

What does cat do?

cat displays the contents of a file in the terminal.

What does cp do?

cp copies a file or directory.

What does mv do?

mv moves a file or directory to another location. It can also be used to rename them.

What does rm do?

rm removes files or directories from the system.

What does ls -l display?

It shows detailed information about files and directories, including permissions, ownership, size, and modification time.

Explain r, w and x permissions.

r (read): allows reading or viewing the contents of a directory or a file.

w (write): allows modifying a file or a directory.

x (execute): allows running a file or accessing a directory.

Explain user, group and others.

User: the owner of the file.

Group: users belonging to the file's assigned group.

Others: everyone else.

What does chmod do?

chmod changes the permissions of a file or directory.

What does chmod 755 represent?

The owner has read, write, and execute permissions, while the group and others have read and execute permissions, where 7—owner, 5—group and 5—others.

What does chmod 644 represent?

The owner has read and write permissions, while the group and others have read-only permission.

What is file ownership?

File ownership identifies which user and group are associated with a file or directory, it affects who can access or modify file or directory.

What is the purpose of /etc?

It contains system-wide configuration files that are used to configure Linux and its services like network.

What is the purpose of /home?

It contains the personal directories and files of regular users.

What is commonly stored under /var?

It usually stores changing data such as log files, caches, databases, and other files generated or updated by system services.

Why should rm be used carefully?

Because rm can permanently delete files, and deleted files may not be easily recoverable.

Why are Linux permissions important?

Permissions control who can read, modify, or execute files and directories. They help protect data and prevent unauthorized access or changes.

How can incorrect permissions create a security risk?

Incorrect permissions can give unauthorized users access to sensitive files or allow them to modify or execute files they should not have access to.

PRACTICAL SCREENSHOTS

mkdir day10_lab · touch notes.txt · ls -l · cat notes.txt · cp command · mv command · chmod u+x demo.sh · chmod 644 notes.txt

```
kali@kali: /day10_lab
Session Actions Edit View Help

(kali@kali)-[/]
$ mkdir day10_lab
mkdir: cannot create directory 'day10_lab': Permission denied

(kali@kali)-[/]
$ sudo mkdir day10_lab

(kali@kali)-[/]
$ touch notes.txt
touch: cannot touch 'notes.txt': Permission denied

(kali@kali)-[/]
$ sudo touch notes.txt

(kali@kali)-[/]
$ ls -l
total 976636
lrwxrwxrwx 1 root root 7 Jun 16 06:14 bin -> usr/bin
drwxr-xr-x 3 root root 4096 Jun 16 10:56 boot
drwxr-xr-x 2 root root 4096 Sep 19 11:19 day10_lab
drwxr-xr-x 18 root root 3240 Sep 19 10:44 dev
drwxr-xr-x 184 root root 12288 Sep 19 10:59 etc
drwxr-xr-x 4 root root 4096 Sep 19 10:58 home
lrwxrwxrwx 1 root root 34 Jun 16 10:56 initrd.img -> boot/initrd.im
g-6.19.14+kali-amd64
lrwxrwxrwx 1 root root 34 Jun 16 10:56 initrd.img.old -> boot/initr
d.img-6.19.14+kali-amd64
lrwxrwxrwx 1 root root 7 Jun 16 06:14 lib -> usr/lib
lrwxrwxrwx 1 root root 9 Jun 16 10:27 lib32 -> usr/lib32
lrwxrwxrwx 1 root root 9 Jun 16 06:14 lib64 -> usr/lib64
drwx----- 2 root root 16384 Jun 16 10:55 lost+found
drwxr-xr-x 2 root root 4096 Jun 16 10:23 media
drwxr-xr-x 2 root root 4096 Jun 16 10:23 mnt
-rw-r--r-- 1 root root 0 Sep 19 11:19 notes.txt
drwxr-xr-x 3 root root 4096 Jun 16 10:27 opt
dr-xr-xr-x 237 root root 0 Sep 19 10:44 proc
drwx----- 4 root root 4096 Sep 19 10:44 root
drwxr-xr-x 41 root root 980 Sep 19 10:58 run
lrwxrwxrwx 1 root root 8 Jun 16 06:14 sbin -> usr/sbin
drwxr-xr-x 3 root root 4096 Jun 16 10:29 srv
-rw----- 1 root root 1000000000 Jun 16 10:56 swap
dr-xr-xr-x 13 root root 0 Sep 19 10:44 sys
drwxrwxrwt 11 root root 300 Sep 19 11:19 tmp
drwxr-xr-x 15 root root 4096 Jun 16 10:27 usr
drwxr-xr-x 12 root root 4096 Sep 12 13:32 var
lrwxrwxrwx 1 root root 31 Jun 16 10:56 vmlinuz -> boot/vmlinuz-6.19
.14+kali-amd64
lrwxrwxrwx 1 root root 31 Jun 16 10:56 vmlinuz.old -> boot/vmlinuz-
6.19.14+kali-amd64

(kali@kali)-[/]
$ cat note.txt
cat: note.txt: No such file or directory

(kali@kali)-[/]
$ cat notes.txt
```

Screenshot taken

View image

```
kali@kali: /day10_lab
Session Actions Edit View Help

(kali@kali)-[/]
$ touch trycp.txt
touch: cannot touch 'trycp.txt': Permission denied

(kali@kali)-[/]
$ sudo touch trycp.txt

(kali@kali)-[/]
$ sudo cp trycp.txt day10_lab

(kali@kali)-[/]
$ sudo mv notes.txt day10_lab

(kali@kali)-[/]
$ touch demo.sh
touch: cannot touch 'demo.sh': Permission denied

(kali@kali)-[/]
$ sudo touch demo.sh

(kali@kali)-[/]
$ mv demo.sh day10_lab
mv: cannot move 'demo.sh' to 'day10_lab/demo.sh': Permission denied

(kali@kali)-[/]
$ sudo mv demo.sh day10_lab

(kali@kali)-[/]
$ sudo chmod u+x demo.sh
chmod: cannot access 'demo.sh': No such file or directory

(kali@kali)-[/]
$ ls day10_lab
demo.sh  notes.txt  trycp.txt

(kali@kali)-[/]
$

(kali@kali)-[/]
$ cd day10_lab

(kali@kali)-[/day10_lab]
$ sudo chmod u+x demo.sh

(kali@kali)-[/day10_lab]
$ sudo chmod 644 notes.txt

(kali@kali)-[/day10_lab]
$ ls -l
total 0
-rwxr--r-- 1 root root 0 Sep 19 11:27 demo.sh
-rw-r--r-- 1 root root 0 Sep 19 11:19 notes.txt
-rw-r--r-- 1 root root 0 Sep 19 11:25 trycp.txt

(kali@kali)-[/day10_lab]
$
```

```
kali@kali: /
Session Actions Edit View Help
(kali@kali)-[~]
$ whpami
Command 'whpami' not found, did you mean:
  command 'whoami' from deb coreutils
Try: sudo apt install <deb name>

(kali@kali)-[~]
$ whoami
kali

(kali@kali)-[~]
$ hostname
kali

(kali@kali)-[~]
$ pwd
/home/kali

(kali@kali)-[~]
$ ls
Desktop  Downloads  Pictures  Public  Videos
Documents Music      Projects  Templates

(kali@kali)-[~]
$ ls -l
total 36
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Desktop
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Documents
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Downloads
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Music
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Pictures
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Projects
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Public
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Templates
drwxr-xr-x 2 kali kali 4096 Sep 12 13:33 Videos

(kali@kali)-[~]
$ ls -la
.          Projects
..         Public
.bash_logout .ssh
.bashrc    Templates
.bashrc.original .vboxclient-clipboard-tty7-control.pid
.cache     .vboxclient-clipboard-tty7-service.pid
.config    .vboxclient-draganddrop-tty7-control.pid
Desktop    .vboxclient-draganddrop-tty7-service.pid
.dmrc      .vboxclient-hostversion-tty7-control.pid
Documents  .vboxclient-seamless-tty7-control.pid
Downloads  .vboxclient-seamless-tty7-service.pid
.face      .vboxclient-vmvga-session-tty7-control.pid
.face.icon .vboxclient-vmvga-session-tty7-service.pid
.gnupg     Videos
.ICEauthority .Xauthority
.java      .xsession-errors
.local     .xsession-errors.old
.Music     .zprofile
.Pictures  .zsh_history
```

